5.24. Problems

Problem 5.24.1. (a) Show that the S_n-representation

$$V'_\lambda := \mathbb{C}[S_n]b_\lambda a_\lambda$$

is isomorphic to V_λ.

Hint: Define S_n-homomorphisms $f : V_\lambda \to V'_\lambda$ and $g : V'_\lambda \to V_\lambda$ by the formulas $f(x) = xa_\lambda$ and $g(y) = yb_\lambda$, and show that they are inverse to each other up to a nonzero scalar.

(b) Let $\phi : \mathbb{C}[S_n] \to \mathbb{C}[S_n]$ be the automorphism sending s to $(-1)^s s$ for any permutation s. Show that ϕ maps any representation V of S_n to $V \otimes \mathbb{C}_-$. Show also that $\phi(\mathbb{C}[S_n]a) = \mathbb{C}[S_n]\phi(a)$, for $a \in \mathbb{C}[S_n]$. Use (a) to deduce that $V_\lambda \otimes \mathbb{C}_- = V_{\lambda^*}$, where λ^* is the conjugate partition to λ, obtained by reflecting the Young diagram of λ.

Problem 5.24.2. Let $R_{k,N}$ be the algebra of polynomials on the space of k-tuples of complex $N \times N$ matrices $X_1, \dots, X_k$, invariant under simultaneous conjugation. An example of an element of $R_{k,N}$ is the function $T_w := \mathrm{Tr}(w(X_1, \dots, X_k))$, where w is any finite word on a k-letter alphabet. Show that $R_{k,N}$ is generated by the elements T_w.

Hint: Consider invariant functions that are of degree d_i in each X_i, and realize this space as a tensor product $\bigotimes_i S^{d_i}(V \otimes V^*)$. Then embed this tensor product into $(V \otimes V^*)^{\otimes n} = \mathrm{End}(V)^{\otimes n}$, and use the Schur-Weyl duality to get the result.

5.25. Representations of $GL_2(\mathbb{F}_q)$

5.25.1. Conjugacy classes in $GL_2(\mathbb{F}_q)$. Let $\mathbb{F}_q$ be a finite field of size q of characteristic other than 2 and $G = GL_2(\mathbb{F}_q)$. Then

$$|G| = (q^2 - 1)(q^2 - q),$$

since the first column of an invertible 2×2 matrix must be nonzero and the second column may not be a multiple of the first one. Factoring,

$$|GL_2(\mathbb{F}_q)| = q(q + 1)(q - 1)^2.$$